Quadcopter UAV Build Guide

A repo-ready build narrative for the completed mini-quad project, focused on assembly, integration, validation, and final configuration capture.

Scope

This guide is written as a practical build reference for anyone using the repository. It documents the assembly order, integrated subsystems, evidence from the final build, and the final configuration artifacts that should be preserved in version control.

Figure 1. Recommended assembly and bring-up sequence used to get from bare hardware to a validated flight stack.

Project summary

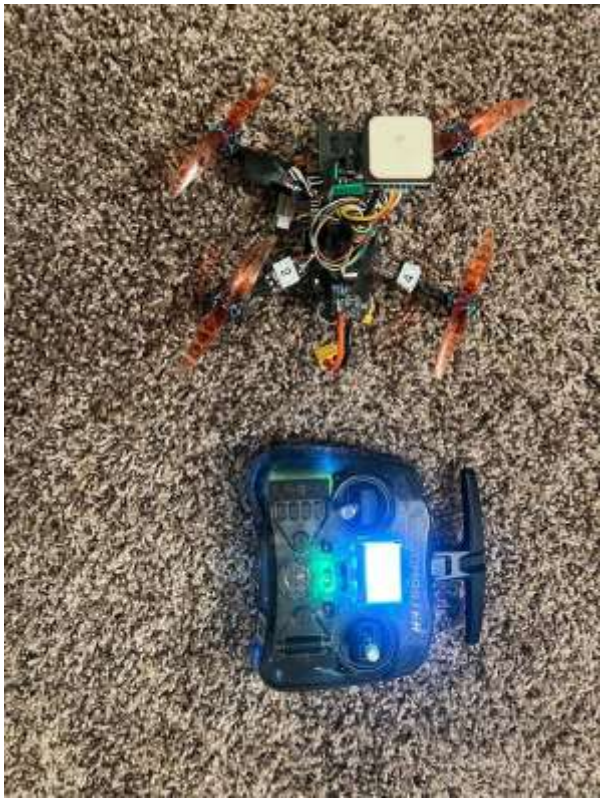
- Built a compact quadcopter UAV around an ArduCopter 4.6.3 flight stack, with a Matek-class flight controller, four brushless motors, four ESCs, onboard LiPo battery power, GPS, ExpressLRS radio control, and optional Wi-Fi telemetry.
- Completed the project as a full integration exercise: mechanical assembly, wiring, power-path soldering, receiver and GPS installation, radio model setup, Mission Planner validation, and preservation of the final parameter file.
- Structured the final deliverables so the repo can support both embedded systems and UAV/robotics audiences: high-level architecture, wiring references, radio setup, Mission Planner checks, and configuration artifacts.

Hardware stack

Subsystem	What was integrated	Why it matters
Airframe	Carbon-fiber mini quad frame, structural arms, standoffs, stack hardware, GPS mast, and battery retention hardware	Provides the mechanical foundation of the UAV and affects packaging, rigidity, mounting stability, and serviceability
Propulsion	4 brushless motors and 4 ESCs connected to the flight controller output and power path	Creates the propulsion chain for lift and maneuvering; *correct ESC setup, motor order, and motor direction are essential for safe operation*
Flight controller	Matek F405-TE / F405-WMN flight controller running ArduCopter 4.6.3	Executes stabilization, arming logic, motor control, serial/UART integration, and *parameterized vehicle configuration*
Radio control link	RadioMaster Pocket transmitter and Happymodel EP2 ExpressLRS receiver	Enables pilot input, switch-based mode control, arming/disarming, and *reliable RF control-link operation*
Navigation / telemetry	GPS module mounted above the frame; *optional DroneBridge ESP32-C3 telemetry module*	Adds navigation data and enables optional Wi-Fi MAVLink telemetry for *monitoring and configuration from a ground station*
Power system	LiPo battery, XT60 connector, heavy-gauge battery leads, ESC wiring, and onboard power distribution	Delivers electrical power to the propulsion and control electronics; wiring quality *and polarity correctness are critical for bring-up*
Configuration toolchain	Mission Planner, EdgeTX, EdgeTX Companion, ExpressLRS Configurator, Backpack firmware, and DroneBridge tools	Supports firmware flashing, parameter setup, radio/*receiver configuration*, telemetry setup, and end-to-end validation

Final build evidence

Top-down view of the integrated quadcopter showing the GPS mast and compact wiring layout.



Completed airframe shown with the configured RadioMaster Pocket transmitter.

Side profile of the final build, showing the stacked electronics and raised GPS placement.

Battery mounting arrangement and underside packaging of the finished frame.

Close-up of the flight-controller area and the power / signal wiring around the stack.

Close-up of the power connector and ESC / wiring region used for final bring-up.

Build sequence in practice

- Assemble the frame first so standoff spacing, stack height, and battery-strap placement are known before electrical packaging begins.
- Mount motors and ESCs before soldering the main battery leads. This keeps motor lead lengths and ESC orientation consistent.
- Complete the XT60 and heavy-gauge battery wiring before mounting the flight controller. Inspect polarity and use strain relief so the battery lead does not twist the board pads.
- Mount the flight controller, then route GPS and receiver wiring while the top of the frame is still accessible. This makes serial/UART wiring cleaner and avoids rework later.
- Flash firmware and configure the radio before propellers are installed. The initial Mission Planner checks should always be done without props.
- Capture the final validated configuration as a .param file and store it in the repo. This gives a clean handoff point for future tuning or rebuilds.

Firmware and tooling baseline

Software / tool	Role in the build	Version used
ArduCopter	Flight-controller firmware	4.6.3
BLHeli_S	Motor-control firmware on ESCs	BLHeli_S
ExpressLRS	Radio-control link	4.0.0
DroneBridge for ESP32	Wi-Fi telemetry bridge	2.1.0
EdgeTX	Radio UI on transmitter	2.11.5
Mission Planner	Ground control and parameter setup	1.3.83
EdgeTX Companion	Radio model configuration	2.11.4
ELRS Backpack	Radio Wi-Fi telemetry feature	1.5.5
ExpressLRS Configurator	Radio / RX flashing utility	1.7.11